```py

# pandas_examples_extracted.py
# ============================
# Ce fichier regroupe les extraits de code Python issus du document fourni
# (Pandas/NumPy). Les exemples sont organisés par sections, avec un minimum
# d'importations redondantes. Certains exemples d'E/S (CSV) sont gardés mais
# protégés par try/except pour éviter des erreurs si les fichiers n'existent pas.

import pandas as pd
import numpy as np

# Si vous souhaitez activer les graphiques, décommentez les lignes ci-dessous :
# import matplotlib.pyplot as plt

# ------------------------------------------------------------------------------
# 1) Structures principales de Pandas
# ------------------------------------------------------------------------------

def example_series():
    s = pd.Series([1, 2, 3], index=["a", "b", "c"])
    print("Series:\n", s)
    return s

def example_dataframe_simple():
    data = {'Name': ['Alice', 'Bob', 'Charlie'], 'Age': [25, 30, 35]}
    df = pd.DataFrame(data)
    print("DataFrame simple:\n", df)
    return df

def example_dataframe_countries():
    countries = ['France', 'Canada', 'Brésil', 'Australie', 'Inde', 'Chine',
    'États-Unis', 'Russie', 'Mexique', 'Japon']
    population = [67.39, 37.59, 211.05, 25.36, 1380, 1441, 331, 146.7, 126.2, 125.8]
    area = [643801, 9984670, 8515767, 7692024, 3287263, 9596961, 9833517, 17098242,
    1964375, 377975]

    sc = pd.DataFrame({
        "Name": countries,
        "Pop": population,
        "Area": area
    })
    print("Countries DataFrame (sc):\n", sc)
    return sc

def example_dataframe_gdp():
    gdp_data = {
        'Name': ['France', 'Canada', 'Brésil', 'Australie', 'Inde', 'États-Unis',
        'Japon'],
        'GDP': [2715, 1643, 2055, 1393, 2875, 21433, 5065]  # PIB en milliards de dollars
    }
    df_gdp = pd.DataFrame(gdp_data)
    print("GDP DataFrame:\n", df_gdp)
    return df_gdp

def example_dataframe_continent():
    continent_data = {
        'Name': ['France', 'Canada', 'Brésil', 'Australie', 'Inde', 'Chine',
        'États-Unis', 'Russie', 'Mexique', 'Japon'],
        'Continent': ['Europe', 'Amérique', 'Amérique', 'Océanie', 'Asie', 'Asie',
        'Amérique', 'Europe', 'Amérique', 'Asie']
    }
    df_continent = pd.DataFrame(continent_data)
    print("Continent DataFrame:\n", df_continent)
    return df_continent

# ------------------------------------------------------------------------------
# 2) Opérations courantes avec Pandas
# ------------------------------------------------------------------------------
```

```
65
66  def example_select_column(df):
67      # Sélection de colonnes
68      age_col = df['Age'] if 'Age' in df.columns else None
69      if age_col is not None:
70          print("Colonne Age:\n", age_col)
71
72  def example_merge():
73      df1 = pd.DataFrame({'Employee': ['Alice', 'Bob'], 'Department': ['HR', 'Finance']})
74      df2 = pd.DataFrame({'Employee': ['Alice', 'Bob'], 'Hire_Date': [2015, 2018]})
75      result = pd.merge(df1, df2, on='Employee')
76      print("Merge df1/df2:\n", result)
77      return result
78
79  def example_io_csv():
80      # Importation/Exportation
81      try:
82          df = pd.read_csv('data.csv')  # Charger un fichier CSV
83          print("CSV head:\n", df.head())
84          # Exporter vers un fichier CSV
85          df.to_csv('output.csv', index=False)
86      except FileNotFoundError:
87          print("[Info] 'data.csv' introuvable - exemple d'E/S ignoré.")
88
89  # -----------------------------------------------------------------------------
90  # 3) Analyse exploratoire des données
91  # -----------------------------------------------------------------------------
92
93  def example_eda(df_like):
94      # Résumé statistique
95      try:
96          print("describe():\n", df_like.describe())
97      except Exception as e:
98          print("[Info] describe() indisponible:", e)
99
100     # Visualisation rapide (désactivée par défaut)
101     # if 'Age' in df_like.columns:
102     #     df_like['Age'].plot(kind='bar')
103     #     plt.show()
104
105     # head/info/describe/shape/columns/dtypes/index (avec garde)
106     try:
107         print("head(3):\n", df_like.head(3))
108     except Exception:
109         pass
110     try:
111         print("info():")
112         df_like.info()
113     except Exception:
114         pass
115     try:
116         print("describe():\n", df_like.describe())
117     except Exception:
118         pass
119     try:
120         print("shape:", getattr(df_like, 'shape', None))
121         print("columns:", getattr(df_like, 'columns', None))
122         print("dtypes:\n", getattr(df_like, 'dtypes', None))
123         print("index:", getattr(df_like, 'index', None))
124     except Exception:
125         pass
126
127 # -----------------------------------------------------------------------------
128 # 4) Exemples concrets
129 # -----------------------------------------------------------------------------
130
131 def example_load_csv_and_explore():
132     try:
133         df = pd.read_csv('data.csv')
```

```
        print(df.head())
        print(df.info())
        return df
    except FileNotFoundError:
        print("[Info] 'data.csv' introuvable - exemple ignoré.")
        return None

def example_filtering():
    data = {'Name': ['Alice', 'Bob', 'Charlie', 'David'],
            'Age': [24, 42, 35, 23],
            'City': ['Paris', 'Londres', 'Paris', 'Berlin']}
    df = pd.DataFrame(data)

    filtered_df = df[df['Age'] > 30]
    print("Age > 30:\n", filtered_df)

    paris_df = df[df['City'] == 'Paris']
    print("City == Paris:\n", paris_df)
    return df, filtered_df, paris_df

def example_stats_descriptives():
    sales_data = {'Product': ['A', 'B', 'C', 'D'],
                  'Sales': [500, 700, 900, 1100],
                  'Profit': [50, 80, 120, 160]}
    df_sales = pd.DataFrame(sales_data)

    print("describe() ventes:\n", df_sales.describe())

    # Ajout / suppression de colonnes
    df_sales['Revenue'] = df_sales['Sales'] * df_sales['Profit']
    print("+ Revenue:\n", df_sales)
    df_sales.drop(columns=['Revenue'], inplace=True)
    print("- Revenue:\n", df_sales)
    return df_sales

def example_merge_again():
    df1 = pd.DataFrame({'Employee': ['Alice', 'Bob'], 'Department': ['HR', 'Finance']})
    df2 = pd.DataFrame({'Employee': ['Alice', 'Bob'], 'Hire_Date': [2015, 2018]})
    merged_df = pd.merge(df1, df2, on='Employee')
    print("Merged again:\n", merged_df)
    return merged_df

def example_groupby():
    df = pd.DataFrame({'Product': ['A', 'B', 'A', 'B', 'A'],
                       'Sales': [500, 700, 600, 800, 400]})
    grouped_df = df.groupby('Product')['Sales'].sum()
    print("Groupby sum:\n", grouped_df)
    return df, grouped_df

def example_apply():
    df = pd.DataFrame({'Sales': [500, 700, 900, 1100]})
    def increase_sales(x):
        return x * 1.1
    df['Increased_Sales'] = df['Sales'].apply(increase_sales)
    print("apply increase 10%:\n", df)
    return df

# ------------------------------------------------------------------------------
# 5) Combinaison NumPy et Pandas
# ------------------------------------------------------------------------------

def example_numpy_combo():
    # DataFrame fictif
    data = {'A': [1, 2, 3, 4, 5], 'B': [10, 20, 30, 40, 50]}
    df = pd.DataFrame(data)

    mean_A = np.mean(df['A'])
    print("Moyenne de A:", mean_A)

```

```
    greater_than_mean = df['A'][df['A'] > mean_A]
    print("A > mean(A):\n", greater_than_mean)

    # Conversions / sélections supplémentaires
    sc = example_dataframe_countries()

    # Sélections
    try:
        print("sc['Pop']:\n", sc['Pop'])
        print("sc[['Name','Pop']]:\n", sc[['Name', 'Pop']])
        print("sc.iloc[0]:\n", sc.iloc[0])
        print("sc.loc[0]:\n", sc.loc[0])
        print("sc[sc['Pop'] > 100]:\n", sc[sc['Pop'] > 100])
    except Exception as e:
        print("[Info] Sélections:", e)

    # Colonnes
    try:
        sc['Densité'] = sc['Pop'] * 1e6 / sc['Area']
        print("+ Densité:\n", sc.head())
        sc['Pop'] = sc['Pop'] * 1.02
        sc = sc.drop(columns=['Densité'])
    except Exception as e:
        print("[Info] Colonnes:", e)

    # Stats
    try:
        print("sum Pop:", sc['Pop'].sum())
        print("mean Area:", sc['Area'].mean())
        print("max Pop:", sc['Pop'].max())
        print("min Area:", sc['Area'].min())
        print("corr:\n", sc[['Pop','Area']].corr())
        print("median Pop:", sc['Pop'].median())
    except Exception as e:
        print("[Info] Stats:", e)

    # Groupby (ajout Continent)
    try:
        sc['Continent'] = ['Europe', 'Amérique', 'Amérique', 'Océanie', 'Asie', 'Asie',
        'Amérique', 'Europe', 'Amérique', 'Asie']
        print("groupby Continent sum Pop:\n", sc.groupby('Continent')['Pop'].sum())
    except Exception as e:
        print("[Info] Groupby:", e)

    # Manipulations
    try:
        print("tri Area desc:\n", sc.sort_values(by='Area', ascending=False).head(3))
        sc['Pop'] = sc['Pop'].fillna(0)
        print("isnull sum:\n", sc.isnull().sum())
    except Exception as e:
        print("[Info] Manipulations:", e)

    # Jointure
    try:
        df_additional = pd.DataFrame({
            'Name': ['France', 'Canada', 'Japon'],
            'GDP': [2715, 1643, 5065]
        })
        sc_with_gdp = pd.merge(sc, df_additional, on='Name', how='left')
        print("sc_with_gdp (left join):\n", sc_with_gdp.head())
    except Exception as e:
        print("[Info] Merge add:", e)

    # NumPy interop
    try:
        pop_np = sc['Pop'].to_numpy()
        pop_np = pop_np * 1.05  # +5%
        print("np.sum(pop_np):", np.sum(pop_np))
        print("np.mean(pop_np):", np.mean(pop_np))
```

```
        print("np.median(pop_np):", np.median(pop_np))
        print("np.std(pop_np):", np.std(pop_np))
        print("np.var(pop_np):", np.var(pop_np))

        sc_np = sc[['Pop', 'Area']].to_numpy()
        print("sc_np.shape:", sc_np.shape)
    except Exception as e:
        print("[Info] NumPy ops:", e)

    return df

# ------------------------------------------------------------------------------
# Main runner (optionnel)
# ------------------------------------------------------------------------------

if __name__ == "__main__":
    s = example_series()
    df_simple = example_dataframe_simple()
    sc = example_dataframe_countries()
    df_gdp = example_dataframe_gdp()
    df_continent = example_dataframe_continent()

    example_select_column(df_simple)
    example_merge()
    example_io_csv()

    example_eda(df_simple if df_simple is not None else sc)

    load_df = example_load_csv_and_explore()
    df_all, df_filtered, df_paris = example_filtering()
    df_sales = example_stats_descriptives()
    example_merge_again()
    df_group, grouped = example_groupby()
    df_apply = example_apply()

    example_numpy_combo()

```